Zinger Market Making on Polymarket

Results to date, capital simulation and per-$100 economics

Research record
Paper trading and public market data only

21 October 2026 (UTC) Supersedes the 19-hour interim edition

Status and risk disclosure.
No order has been placed and no money spent. Every figure is a simulation on live public data. On the full 24-hour recording, quote settings chosen on the first half **lost $42.44 per day on about $13,000 (−0.33% per day) on the unseen second half**. The 19-hour interim showed +0.14% per day; that gain did not hold. Most markets made a little; three large adverse moves outweighed them. Real capital remains conditional on the gates in Section 12.

1 Summary

- **Reward capture works mechanically.** After fixing churn and stale-book handling, the runner earns a projected reward share in every market it quotes. Liquid markets paid a steady $10–14 per day at minimum size across 40 markets.
- **Fill losses decide the result.** Over the full day, adverse fills cost more than the reward at every quote distance when one setting is used for all markets.
- **Split-sample test, full day.** Per-market settings chosen on the first half returned −0.33% per day on the second half for liquid markets and −2.77% for thin books. Capping size at the reward minimum gave the same rate; losses scale with size.
- **Losses are concentrated.** Three liquid markets lost $71.44 per day between them; the other 37 made +$29.00. The median market made +$0.53 per day.
- **Capital simulation.** Small books ($100–$1,000, 2–9 markets) were positive on the second half; books of $2,500 and above lost about 0.3% per day.
- **Expectation.** No positive edge is demonstrated. The profile is small steady gains with occasional large losses, which needs news and jump protection before any further sizing.

2 What was built

Component	Purpose
Paper market-making runner	Quotes against live books; simulates queue position, fills, inventory, fees and recovery; zero orders
Resting quotes	Orders keep their queue position while the target stays within one tick
Connection-health freshness	A book counts as current while its websocket answers a one-second heartbeat
Reward accrual	Polymarket's quadratic reward score, sampled each minute, kept separate from cash
Market scanner	Every reward-eligible market from the CLOB, ranked by projected reward per $100
Recorder	One snapshot per minute of every book level inside the reward band, plus trades
Optimiser and split test	Sweeps quote distance and size per market with a pessimistic fill cost; tests settings on unseen data
Capital simulator	Allocates a budget across markets in first-half rank order and scores it on the second half

Table 1. Components built during the research.

3 Execution results from the paper runner

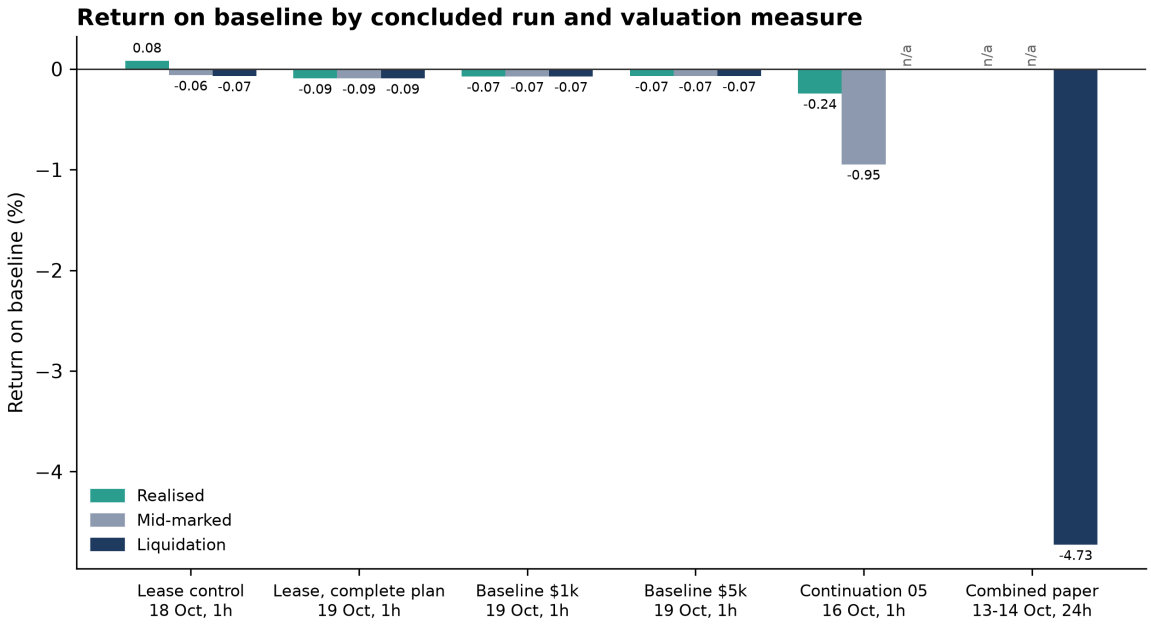


Figure 1. Return on baseline for each concluded paper run (1–25 fills per run).

Run	Baseline	Fills	Liquidation P/L	Return
Lease control, 18 Oct, 1 h	\$1,000	1	−\$0.70	−0.07%
Lease, complete plan, 19 Oct, 1 h	\$1,000	2	−\$0.94	−0.09%
Baseline \$1k, 19 Oct, 1 h	\$1,000	2	−\$0.74	−0.07%
Baseline \$5k, 19 Oct, 1 h	\$5,000	2	−\$3.51	−0.07%
Continuation 05, 16 Oct, 1 h	\$1,000	25	not verified	realised −0.24%
Combined paper, 13–14 Oct, 24 h	\$2,000	—	−\$94.52	−4.73%

Table 2. Concluded paper runs. Total simulated MM volume was about \$92.

4 What broke, and what was fixed

Problem	Effect	Fix
Startup wait with no deadline	Three captures ran 17 hours with no data	Startup deadline, data watchdog, hard stop, one capture at a time
1.5 s REST freshness rule	Books stale 72% of the time	Websocket heartbeat freshness; stale share 0.7%
Quotes expiring every 10 s	Constant loss of queue position	Resting quotes with a one-tick tolerance
Trade endpoint ignores its window	Wrong volume, missed fills	Cursor paging filtered to each sample window
Snapshots kept the far end of the book	Early recordings unusable for re-scoring	Levels selected by distance from the midpoint
Single-snapshot reward estimates	\$8,752/day on thin books	Competition floor, then a 24-hour recording
Minute-cadence stub quotes	+\$70/day mostly from one news shock and one live game	Stub used only for reward share

Table 3. Defects found and corrected.

5 Benchmark against an open-source market maker

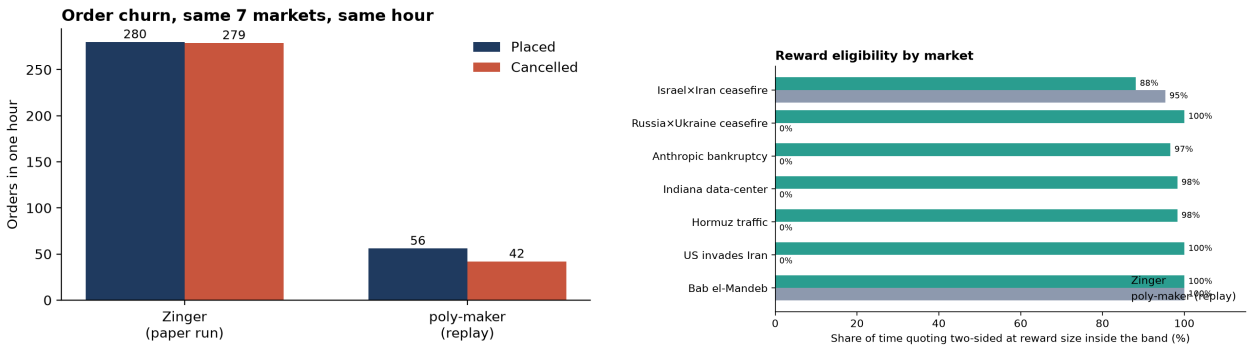


Figure 2. Same 7 markets, same hour. Left: orders placed and cancelled. Right: time quoting two-sided at reward size inside the band.

Zinger churned five times more than `poly-maker`'s logic but qualified for rewards in all 7 markets; `poly-maker`'s copied profile under-sized its NO bids in 5. `poly-maker`'s own notes report a live loss of \$15.51 from one oversized fill, the same failure mode seen here.

6 Market reach

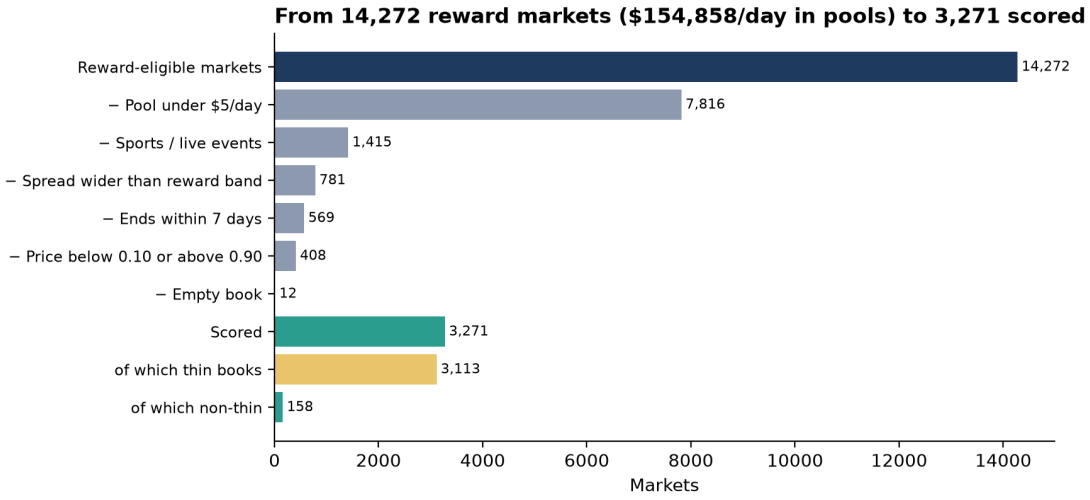


Figure 3. 14,272 reward markets ($154,858 per day in pools) narrowed to 3,271 scored: 158 liquid, 3,113 thin.

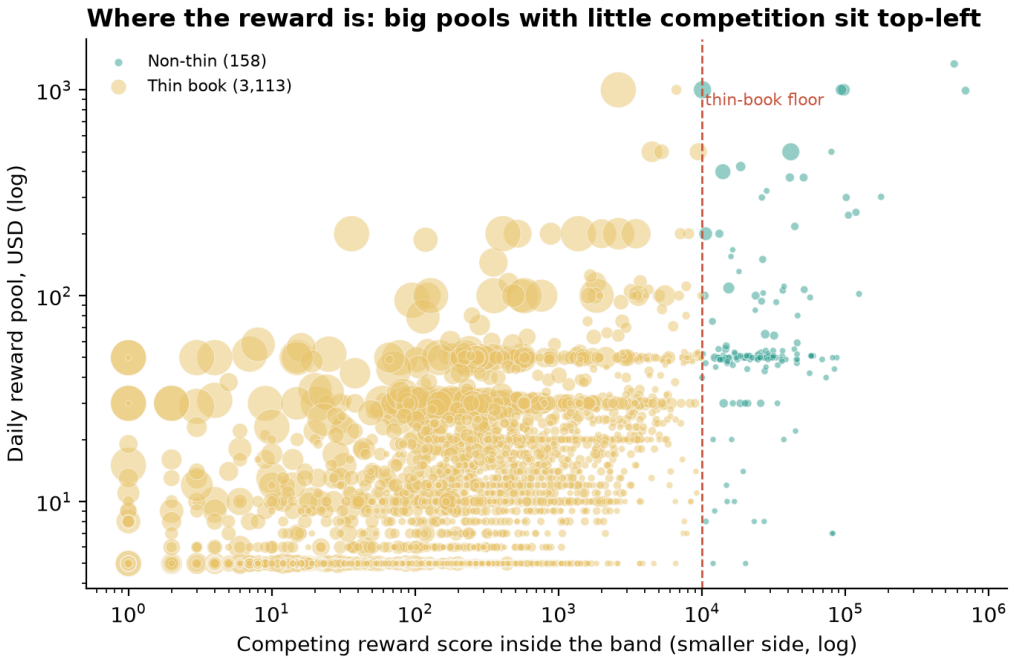


Figure 4. Daily pool against competing reward score.

7 24-hour recording

80 markets (40 liquid, 40 thin) were recorded once a minute from 20 October 18:40 UTC to 21 October 18:40 UTC: 1,440 minutes, 105,449 samples, no errors.

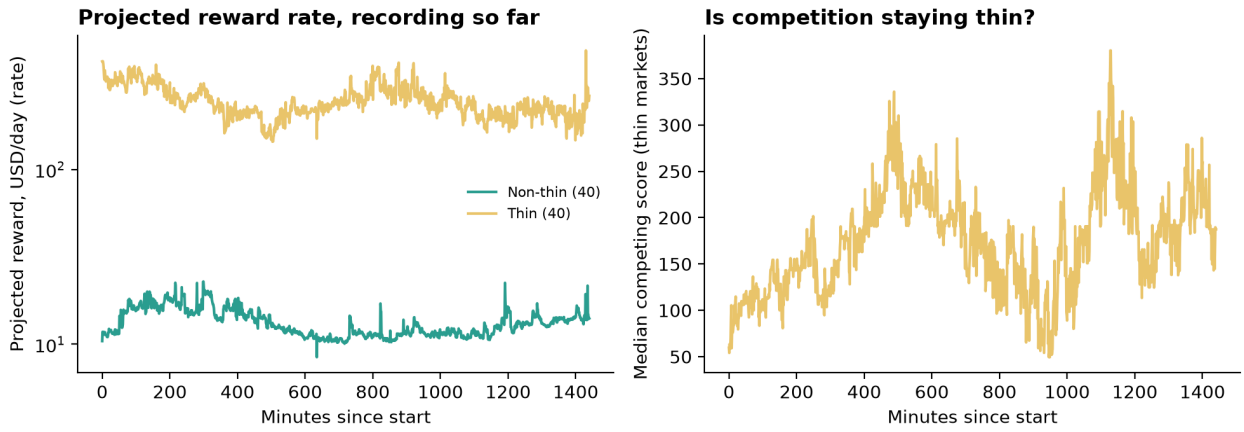


Figure 5. Left: projected reward rate by group. Right: median competing score on thin markets.

	First minute	Last minute	Average
Liquid projected reward	$10.4/day	$14.1/day	$13.4/day
Thin projected reward	$414/day	$263/day	$246/day
Thin median competing score	60	187	—

Table 4. Reward before fill costs. Liquid rewards were stable; thin rewards fell as competition arrived.

8 Parameter tests

For each market-minute the optimiser tries 0–3 ticks back from the touch at 1, 2 and 4 times the reward minimum. Reward is scored against the recorded book. Fill cost is pessimistic: a fill counts only when the price moves through the quote and is marked 10 minutes later; benign fills that would earn spread are ignored.

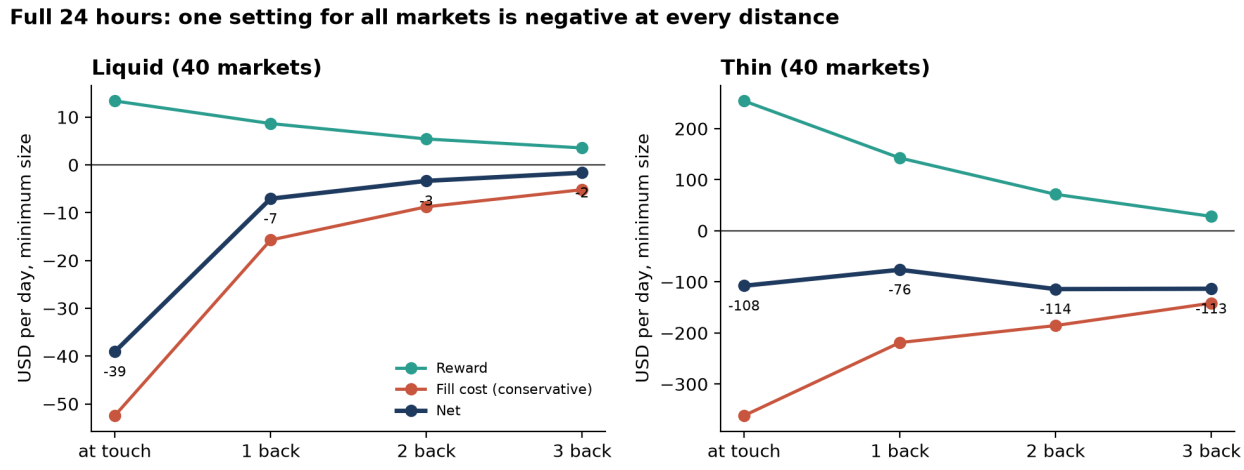


Figure 6. Reward, fill cost and net by quote distance at minimum size, full 24 hours.

Liquid, one setting	Reward	Fill cost	Net	Net %/day	Fills/day
At the touch	\$13.37	−\$52.45	−\$39.08	−1.13%	354
1 tick back	\$8.65	−\$15.72	−\$7.07	−0.21%	36
2 ticks back	\$5.43	−\$8.77	−\$3.34	−0.10%	20
3 ticks back	\$3.55	−\$5.19	−\$1.64	−0.05%	9

Table 5. Thin markets were negative at every distance (−\$76 to −\$114 per day).

8.1 More interactions lower returns

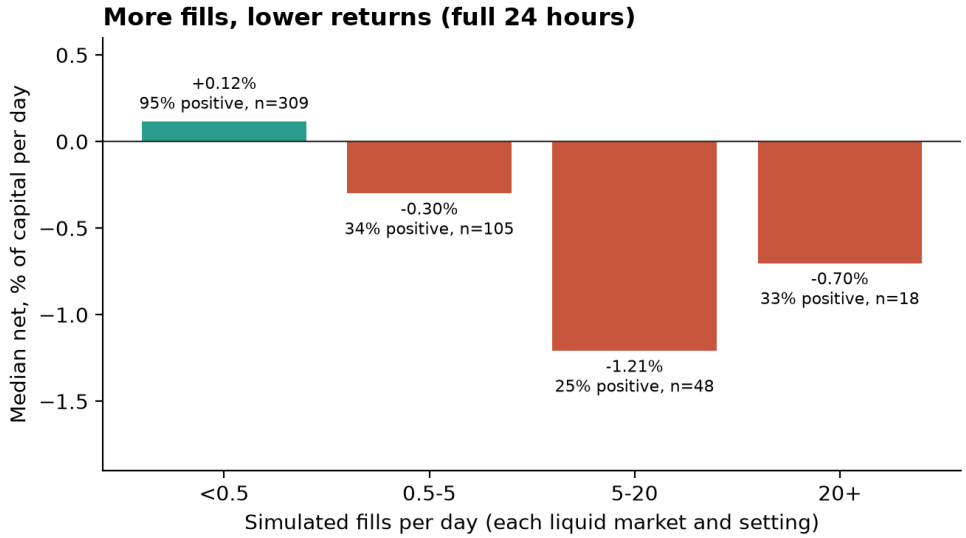


Figure 7. Every liquid market and setting, grouped by simulated fills per day.

Settings with almost no fills earned a median +0.12% per day (95% positive); 5–20 fills per day earned a median −1.21%. Quoting faster or closer increases fills and lowers returns.

8.2 Split-sample test

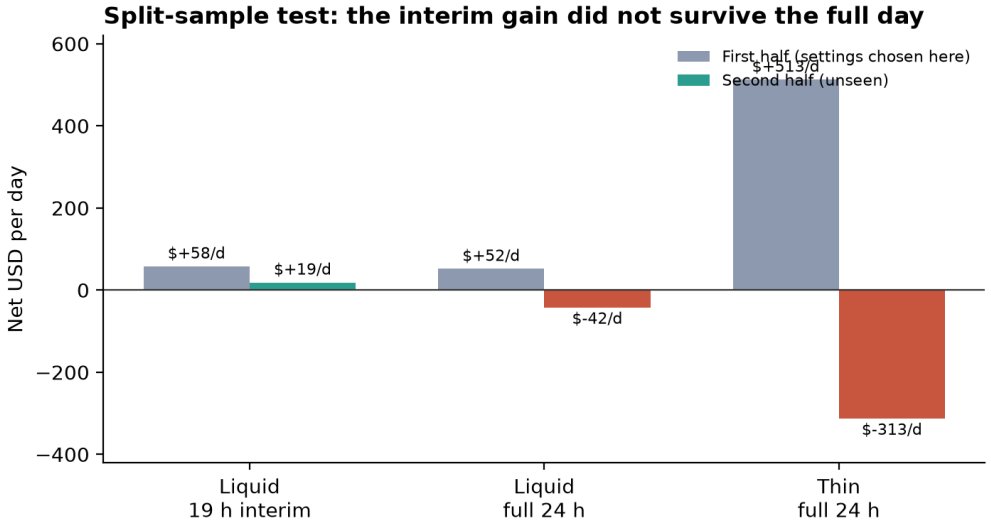


Figure 8. Settings chosen per market on the first half and scored on the unseen second half.

Test	First half	Second half	Capital	Rate
Liquid, per-market settings (19 h interim)	+$57.83	+$18.54	$12,982	+0.14%
Liquid, per-market settings (full day)	+$51.83	−$42.44	$12,975	−0.33%
Liquid, size capped at minimum	+$13.21	−$10.44	$3,183	−0.33%
Liquid, one group setting (3 back, 4×)	+$4.18	−$17.87	$13,095	−0.14%
Thin, per-market settings (full day)	+$512.69	−$312.91	$11,308	−2.77%

Table 6. Net USD per day. 29 of 40 liquid markets were positive in the second half; three lost $71.44 between them.

9 Capital simulation

For each budget, markets that were positive in the first half are funded in rank order at minimum size, then scaled to 2× and 4× while budget remains. The allocation is scored on the unseen second half.

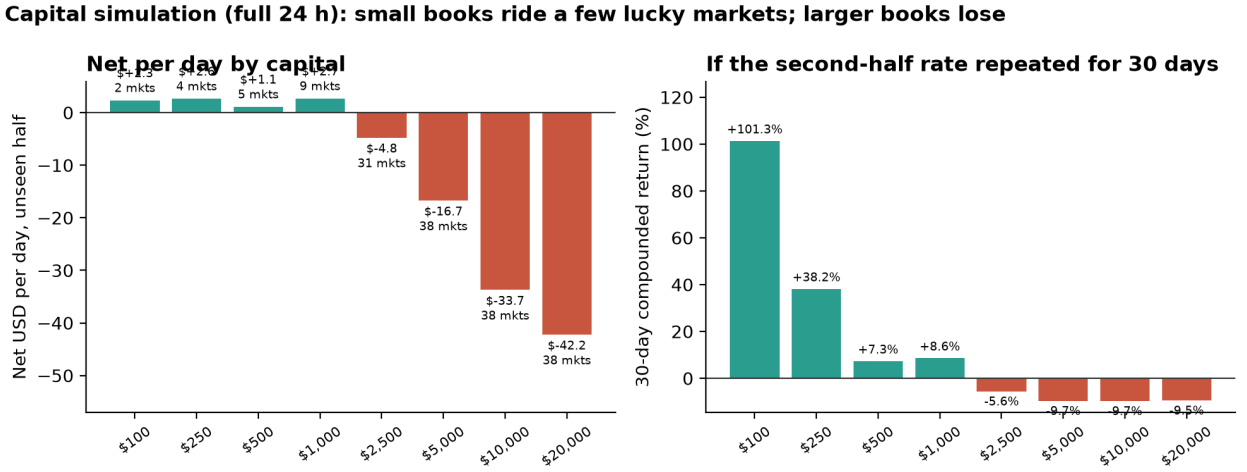


Figure 9. Net per day and 30-day compounded return by capital, full 24-hour recording.

Capital	Deployed	Markets	Net $/day	Rate/day	30 days, compounded
$100	$99	2	+$2.33	+2.36%	+101% (+$100)
$250	$244	4	+$2.65	+1.08%	+38% (+$93)
$500	$471	5	+$1.11	+0.24%	+7% (+$34)
$1,000	$970	9	+$2.67	+0.28%	+9% (+$83)
$2,500	$2,494	31	−$4.77	−0.19%	−6% (−$139)
$5,000	$4,955	38	−$16.75	−0.34%	−10% (−$479)
$10,000	$9,910	38	−$33.74	−0.34%	−10% (−$964)
$20,000	$12,730	38	−$42.24	−0.33%	−9% (−$1,208)

Table 7. Small budgets fund only the 2–9 top-ranked markets, which happened to do well on the second half; this is a small sample and not evidence of a repeatable edge. Above $2,500 the result converges on the portfolio rate of about −0.33% per day. Compounded figures assume the rate repeats for 30 days.

10 Per-$100 economics

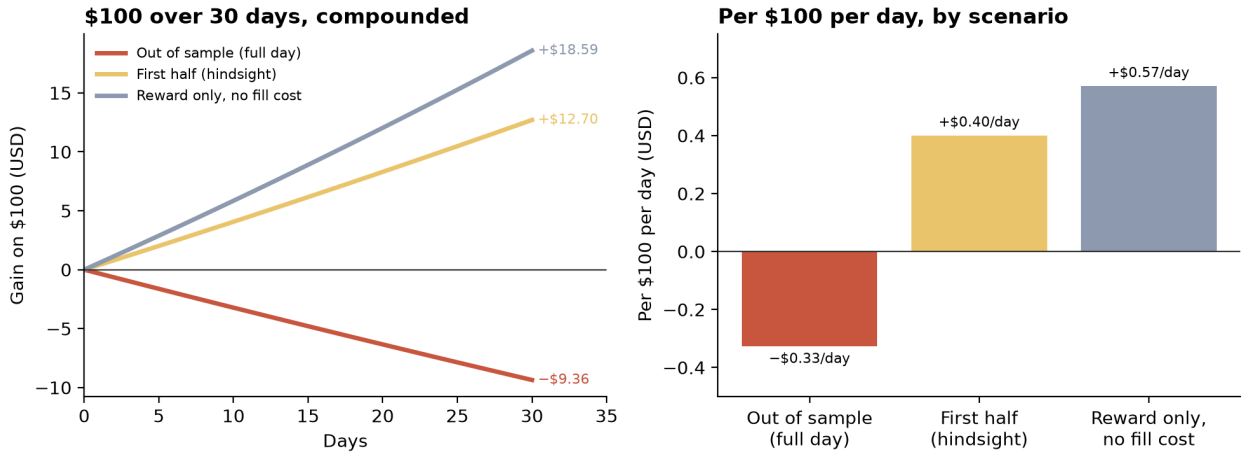


Figure 10. Per $100 under three scenarios.

Scenario	Per $100/day	30 days	Basis
Out of sample (full day)	−$0.33	−$9.33	Settings tested on unseen data; the figure to plan on
First half (hindsight)	+$0.40	+$12.70	Settings chosen and scored on the same data
Reward only, no fill cost	+$0.57	+$18.60	Best 10 liquid markets, snapshot, ignoring fills

Table 8. Hindsight and reward-only figures are upper bounds, not forecasts.

11 Risks

- **Tail losses from adverse moves.** The decisive cost: a few news-driven moves outweighed a day of reward.
- **Overfitting.** Settings that looked best on one half lost on the other; the 19-hour interim overstated the result.
- **Thin and manipulable books.** Projected rewards there did not survive fills.
- **Concentration and capacity.** Rewards come from fixed pools ($5,792 per day across the 40 liquid markets), so returns fall as size grows.
- **Model gaps.** Rewards are projected, queue position and routing are simplified, and benign fills are ignored.
- **Multi-outcome support and access.** Most of the best markets are multi-outcome, which the runner cannot quote; eligibility from the operating location is unverified.

12 Gates before real capital

Current status: none of these gates has passed. Each is a condition for a capital decision.

1. Positive net out of sample across several days and splits, with tail losses included.

2. Jump and news protection that pulls quotes before large moves, tested on recorded data.
3. The same result from the full paper runner (queue, fills, inventory), not only the optimiser.
4. Multi-outcome quoting implemented and tested; independent auditor updated.
5. Live-path readiness, eligibility and wallet checks.
6. Written sign-off on a canary size and stop rules.

13 Next steps

1. Add a jump filter to the optimiser (pull quotes when the midpoint moves more than a set amount, or volume spikes) and rerun the split test.
2. Record further days and repeat the split test on each, including a test on a different day than the one used to choose settings.
3. Add multi-outcome quoting to the paper runner and run it on the liquid markets.

Reproducibility

- Scan: `scripts/mm-reward-scan.mts`; recording: `scripts/mm-reward-projection.mts`, run `2026-10-20T18-40-36-48`
- Optimiser: `scripts/mm_param_optimiser.py`; split test and capital simulation: `scripts/mm_walkforward_capital.py` outputs in `final-20261021/`.
- Benchmark and fixes: `docs/MM_POLYMAKER_BENCHMARK_2026-10-20.md`, `docs/MM_RESET_2026-10-20.md`, `docs/MM_REWARD_PROJECTION.md`.